SAGAR PUBLIC SCHOOL

Integrated Programme • Topic: Work, Power & Energy

Practice Sheet – 3 : ANSWER KEY

Q	Section	Source	Answer	Remark
1	Energy & Conservation	PYQ JEE Main 2017, 8 Apr M	(A) $3h$	
2	Energy & Conservation	Module S-1 Q25	$\frac{2m_1^2g^2}{k}$	
3	Energy & Conservation	PYQ JEE Main 2023, 6 Apr E	375	
4	Energy & Conservation	Module O-1 Q18	(D) 80 m/s	
5	Energy & Conservation	PYQ JEE Main 2024, 30 Jan M	(C) $\sqrt{10}$ m/s	
6	Energy & Conservation	Module S-2 Q7	$4a/3$	
7	Energy & Conservation	PYQ JEE Main 2026, 23 Jan M	(D) $3E$	
8	Energy & Conservation	PYQ JEE Main 2022, 26 Jul M	(B) $v/2$	
9	Work–Energy Thm	PYQ JEE Main 2026, 6 Apr M	(D) -9.98 J	
10	Work–Energy Thm	Module S-1 Q13	400 J	
11	Work–Energy Thm	PYQ JEE Main 2016, 10 Apr M	(C) -4500 J	
12	Work–Energy Thm	Module O-1 Q10	(A) 1.17 m	
13	Work–Energy Thm	PYQ JEE Main 2024, 9 Apr E	58 J	
14	Work–Energy Thm	PYQ JEE Main 2019, 9 Jan E	(D) 900 J	
15	Power	PYQ JEE Main 2023, 13 Apr M	(A) 16	
16	Power	Module S-2 Q11	30 m/s	
17	Power	PYQ JEE Main 2025, 29 Jan E	(D) $P^2 \propto v^5$	
18	Vertical Circular	PYQ JEE Main 2024, 4 Apr E	(D) 21.9 m/s	
19	Vertical Circular	Module O-1 Q29	(A) $g\sqrt{10}$	
20	Vertical Circular	Module JA Q12	(D) radially inwards initially, outwards later	

Answers are taken from the answer keys printed in the source files (PYQ key / module key). Note: the module key lists a few answers in different notation; numerical answers are in the units stated in the question.